

MSDS 535 – Week 13

Chapter 12, Sections 1 and 2

Overview and Objectives

- Discuss Data Mining trends and research frontiers.
- Interpret Mining rich data types.
- Develop Data mining applications.

Mining text data, Data Formats

- Data can be structured (relational databases), semistructured (XML, JSON, HTML), or unstructured (like text from various sources).
- Unstructured data, such as text from books, emails, social media, constitutes over 80% of world data.
- Structured data is organized, while unstructured data lacks a predefined structure.

Text Mining Overview

- Text mining involves extracting high-quality information, structures, and patterns from text sources.
- Tasks in text mining include concept/entity extraction, sentiment analysis, and document summarization.
- Methods for text mining tasks include supervised, semisupervised, distantly supervised, weakly supervised, and self-supervised approaches.

Representation Learning for Text

- Early approaches used symbol-based representations like one-hot vectors and bag-of-words models.
- Recent studies favor distributed representation, creating low-dimensional vectors with deep neural networks.
- Word embedding methods (Word2vec, GloVe, fastText) encode words in continuous vector spaces for semantic meaning.

Pretrained Language Models (PLMs)

- Models like ELMo, BERT, RoBERTa capture contextualized word embeddings from larger corpora.
- These PLMs encode linguistic knowledge within multilayer networks, achieving state-of-the-art results in NLP.
- They follow a pretraining-and-fine-tuning pipeline, leveraging the same structure for initialization and task-specific learning.

Text Mining Tasks

- Information Extraction (IE) includes NER, coreference resolution, relationship extraction, and event extraction.
- Taxonomy Construction uses text analysis to automatically generate structured classifications.
- Text Clustering & Topic Modeling aim to group texts by similarity or discover abstract topics within documents.

Recent Developments & Research

- PLMs have evolved from ELMo and BERT to newer models like RoBERTa, XLNet, and ELECTRA.
- Representation learning shifted from symbol-based to distributed representations for better semantic understanding.
- Word Embedding methods expanded from Word2vec to include GloVe, fastText, and even Poincaré embeddings in hyperbolic space.

Importance of Space and Time

- Space and time hold crucial roles in various domains like smart cities, epidemiology, and earth sciences.
- Existing data mining principles can be tailored to suit spatial-temporal data applications.
- Adapting these techniques requires addressing unique challenges and opportunities posed by spatial-temporal data characteristics.

Properties of Spatial-Temporal Data

- Dependencies often occur in proximate locations and times, termed as auto-correlation, impacting analyses like traffic statuses in adjacent intersections.
- Spatial-temporal data often exhibit nonstationary distributions and periodic patterns, posing modeling challenges while offering analytical opportunities.

Spatial and Temporal Data Types

- Events represent occurrences at specific spatial and temporal coordinates, aiding in scenario analyses like traffic accidents.
- Trajectories track object paths over space and time, essential in transportation and ecological studies for relationship assessments.
- Point reference and raster data serve diverse purposes from reconstructing fields to remote sensing and medical imaging.

Models for Spatial and Temporal Data

- Spatial models like object, field, and spatial network models offer different perspectives on spatial elements and their interconnections.
- Temporal models, including snapshot, change, and event/process models, capture temporal aspects and spatial changes over time.

Adaptation of Data Mining Methods

- Tailoring existing data mining techniques for spatial-temporal applications is crucial.
- Developing methodologies focused on mining unique spatial and temporal relationships is vital for comprehensive analyses.

Additional Resources

- Surveys like those by Atluri, Karpatne, and Kumar and Shekhar, Vatsavai, and Celik provide comprehensive insights into mining spatial-temporal data.

The World of Interconnectedness

- Many complex systems, like online social platforms and biological networks, are represented as graphs.
- These graphs consist of nodes (system objects) connected by edges (interactions among objects).
- Such representations capture interactions in various domains, spanning from online user networks to molecular interactions.

Graph Categorization

- Graphs can be plain or attributed based on node and edge attributes.
- They may be static or dynamic, evolving over time, and homogeneous or heterogeneous in terms of node and edge types.
- Examples illustrate homogeneous (collaboration network) and heterogeneous (bibliographic network) networks.

Insight Discovery in Graphs

- Graph modeling explores the generation mechanisms behind real-world graphs like ER models and preferential attachment models.
- Realistic graph generators mimic properties of real-world graphs; deep graph generative models learn to replicate various graph types.
- Graph neural networks aid in generating synthetic graphs but might lack interpretability and focus on homogeneous networks.

Heterogeneous Network Mining

- Heterogeneous networks offer rich semantics through metapaths connecting different node types.
- Tasks like similarity search, classification, clustering, and embedding benefit from diverse metapaths.
- Existing methods often require predefined metapaths, suggesting the need for automated inference from input data.

Knowledge Graph Mining

- Knowledge graphs link entities through human-interpretable semantics, enabling applications like question answering and fact checking.
- Construction methods, including extractive methods and language model-based approaches, aim to automate knowledge graph creation.
- Challenges persist, such as limited explicit information extraction and the need for dynamic, high-precision updates.

References and Contributions

- Various graph models, from ER graphs to deep generative models, have contributed to understanding and simulating real-world graphs.
- Methods in heterogeneous network mining rely on predefined metapaths, urging exploration in automated inference.
- Knowledge graph construction methods aim to automate the process, yet challenges in scaling and precision remain.

Understanding Sentiments and Opinions

- Sentiments represent emotions toward a situation or event, while opinions are judgments often not based on factual knowledge.
- Sentiments are feelings expressed, whereas opinions are views formed about something, sometimes subtly different yet highly related.

Techniques in Sentiment Analysis and Opinion Mining

- This interdisciplinary field utilizes natural language processing and often frames problems as classification tasks.
- Techniques range from sentiment lexicons for classification to clustering for grouping users by opinions.
- Understanding sentiments and opinions requires leveraging insights from psychology and social sciences.

Dynamic Nature and Technical Directions

- The field continually adapts to new media types, adopting the latest data mining techniques.
- Exploration of sentiment and opinion analysis occurs at different granularities, from document-level to aspect-level analysis.
- Lexicon-based sentiment classification faces challenges due to subtleties in language and domain-specific contexts.

Analysis of Conversational Content

- Conversational content presents more challenges due to interactive and interdependent nature; new sentiments like agreements and disagreements emerge.
- Conducting data mining tasks in conversations includes sentiment consistency checks and identifying groups sharing similar opinions.

Intent Mining and Ethics

- Intent mining aligns with sentiment analysis, identifying sentiments within sentences expressing aspirations or desires.
- Maintaining social good involves detecting opinion spamming and assessing review quality, often requiring network mining techniques.

Applications of Sentiment Analysis and Opinion Mining

- Applications range from aiding purchase decisions in e-commerce to triggering automated actions in stock market prediction.
- Sentiment analysis finds utility in government administration, monitoring public sentiments and opinions for various purposes.
- Challenges in these applications include scalability, accuracy, robustness, fairness, and privacy protection.

Historical and Informative References

- Sentiment analysis and opinion mining have seen substantial interdisciplinary growth since the early 2000s, explored in numerous publications and survey books.

Truth Discovery

- Involves assessing the true value for a data item when multiple sources present conflicting information.
- Divided into single-truth (one true value) and multitrueth (multiple true values) scenarios.
- Challenges include voting methods, unreliable sources, and the need to consider source reliability.

Source Reliability Assessment

- Supervised learning for source reliability might not be practical due to the vast number of sources.
- Realistic solutions involve weakly supervised or self-supervised methods based on extensive unlabeled data.

Iterative Truth Discovery

- Mutual enhancement between source trustworthiness and claim reliability is essential.
- The process involves iteratively estimating source trustworthiness and true values until convergence.

Misinformation Identification

- Different from truth discovery, this involves identifying deliberately false or misleading information.
- Misinformation, disinformation, and fake news propagate fear and suspicion, often spread through social media.

Detection Techniques

- Methods akin to truth discovery, like mutual enhancement between source trustworthiness and claim credibility, can assist in identifying misinformation.
- Identifying credible information involves considering consistency from trustworthy sources and flagging conflicting information.

Challenges and Solutions

- Distinguishing misinformation from genuine information on social media requires measures beyond simple votes.
- Building information literacy through education, critical thinking, and algorithms for fact-checking are frontline strategies.

Evolving Research

- Ongoing research involves natural language processing, social network analysis, and algorithmic fact-checkers.
- Historical studies trace the origins of misinformation while recent research focuses on detection and handling.

References

- Notable studies include approaches like TruthFinder, studies on structured data in the deep web, and research on truth finding under multitrueth assumptions.

Information and Disease Propagation

- Data mining plays a crucial role in comprehending information and disease spread through extensive digital data trace, given the unprecedented scale and speed.
- Research in this domain segregates into prediction problems (classification, publication timing, granularities) and optimization problems (influence maximization, source localization, activity shaping, graph connectivity).

Applications of Propagation

- In social media, it aids in predicting virality, identifying rumor sources, and halting misinformation dissemination effectively.
- Computational epidemiology leverages these models to uncover critical network conditions necessary for the outbreak of diseases.

Prediction Problems of Propagation

- Classification and regression methods are employed to predict information popularity using defined measures.
- Predictions made before or after publication provide varying levels of reliability based on the stage of information dissemination observed.
- Forecasting at different granularities allows simultaneous predictions on individual items/users or clusters.

Optimization Problems of Propagation

- Influence maximization focuses on maximizing infected nodes by selecting an optimal set of initial nodes.
- Source localization aims to identify the source nodes by observing partial cascades, which can be challenging with incomplete observations.
- Activity shaping manipulates user behavior with varying incentives and objectives, unlike influence maximization's fixed incentives.
- Graph connectivity optimization alters graph topology to influence information propagation outcomes.

Models for Information Diffusion Problems

- Feature-based models utilize supervised data mining with various features for prediction.
- Generative models, like epidemic models and stochastic processes, capture various information diffusion processes.
- Deep learning-based models, flexible and assumption-free, incorporate multimodal data effectively.
- Computational epidemiology models, including SIR and SEIR variants, simulate disease spread.

Current Challenges and Future Directions

- Pandemic modeling focuses on predicting transmission rates and assessing the impact of government policies.
- Challenges include identifying sources with minimal cascades and interpreting complex shaping behaviors.
- Future exploration targets understanding the predictability and interpretability of propagation models.

Notable References

- Kempe, Kleinberg, and Tardos' work on influence maximization and related studies offer valuable insights.
- Farajtabar et al.'s strategies for activity shaping and recent pandemic modeling research provide essential contributions.
- Zhou et al.'s survey comprehensively covers various prediction models for information diffusion.

Productivity and Team Science

- An increasing interest surrounds team performance, member productivity, and the impact of team-produced content in recent years.
- Teams, characterized by collaborative work toward specific goals, often function within hierarchical structures in modern organizations and operate within underlying networks like communication or social networks.

Types of Teams in Real-World Scenarios

- Film production teams and sports teams illustrate diverse examples where individuals with various roles collaborate to achieve a common objective.
- These teams can be embedded within networks, like actors connected based on shared movie participation or research teams structured by coauthorship and expertise.

Key Questions in Team Science Answered by Data Mining

- Team Performance Characterization investigates patterns distinguishing high-performing teams from struggling ones, influenced by team members, environments, and tasks.
- Team Performance Prediction focuses on forecasting team performance, the impact of team-produced content, and the nonlinear correlation between team performance and member productivity.

Optimizing Team Performance

- Team leaders often optimize team composition, considering not only skills but also network connectivity, akin to player rotations in sports.
- Scenarios like downsizing, expansion, conflict resolution, and real-time optimization pose unique challenges in maintaining high-performing teams.

Explaining Team Performance

- Limited literature delves into providing intuitive explanations for team performance predictions or optimization outcomes.
- Sparse methods utilize influence functions to interpret performance predictions or identify influential network elements in optimization scenarios.

Notable References

- Hackman and Katz's formal introduction of team definitions and various real-world team examples.
- Studies by Uzzi, Mukherjee, Stringer, Jones, Li, Tong, Tang, Fan, Zhou, Li, and Tong offer insights into team performance prediction, trajectory forecasting, and optimization, exploring complexities like network dynamics and domain heterogeneity.